

CONSTRUCTION OS

PRD - QA/QC Dashboard

QA/QC Manager Role Workspace

Product purpose

Define the exact QA/QC Manager role shell, fixed dashboard, field-quality control surfaces, NCR/CAPA management, quality gates and handover readiness behavior. The experience is exception-first: know what must be inspected, what failed, what is blocking work, what requires verification and what prevents handover.

Field	Value
Document	PRD_QAQC_Dashboard
Version	1.0
Status	Implementation-ready role dashboard specification
Primary role	QA/QC Manager / Project QA/QC Manager
Base requirements	PRD_QAQC_Handover + QA/QC & Handover Full Architecture + Complete Role & Permission Matrix + Approvals & Workflow Engine
Connected domains	Projects, Work Progress, Tasks, Documents, Contracts, Procurement, Inventory, Assets, Units, Calendar, HSE, Government & Legal, Finance, Workflow
Scope	QA/QC role dashboard + exact sidebar + quality-control and handover management workspaces
Date	2026-08-07

Permanent dashboard scope lock

The QA/QC Dashboard and QA/QC role sidebar shall contain only the elements explicitly defined in this PRD. Developers must not add, infer, duplicate, substitute or auto-expose extra dashboard widgets, modules or shortcuts without an approved PRD revision. Platform Configuration may hide approved elements; it may not create unapproved ones.

1. Document Authority, Precedence and Product Position

Priority	Authority	Controls
1	Complete Role & Permission Matrix	QA/QC authority, project/record scope, protected fields, deny-by-default behavior, separation of duties, revocation and audit.
2	QA/QC & Handover Full Architecture v1.0	Quality planning, ITPs, inspections, tests, releases, NCR/CAPA, defects/snags/punch, gates, commissioning, handover, DLP, data model, APIs and source boundaries.
3	PRD_QAQC_Handover v1.0	Operational dashboard metrics, mobile/offline rules, permission hooks, platform configuration and quality/handover lifecycle requirements.
4	Approvals & Workflow Engine v1.0	Decision routing, approval matrices, participants, work items, delegation, SLA, SoD and source finalization.
5	Role Context and Shared Module Navigation	QA/QC shell remains active when authorized Project, Task, Document, Inventory, Asset, Unit, Contract, Calendar or Workflow content opens.
6	Platform UI/UX Architecture	Shared shell, Role Center, Work Inbox, Quick Create, progressive disclosure, responsive behavior, accessibility and motion.
7	This PRD	Exact QA/QC Manager dashboard composition, sidebar, project quality workspace and role-specific UX decisions.

1.1 Approval dependency resolution

Resolved dependency

The QA/QC source PRDs describe organization-wide approvals as future or capability-hook scope. The platform Approvals & Workflow Engine now exists and governs routing, participants, decisions, delegation, SLA and escalation. QA/QC and Handover still own their source records, quality results, readiness states and final business transitions.

1.2 Executive purpose

- Give QA/QC Managers one calm control surface for inspections, quality failures, corrective verification, blocking gates and handover risk.
- Prioritize actionable quality risk before passive analytics.
- Keep company, project, contractor, discipline, location and date context explicit.
- Make inspection, NCR, defect and verification work reachable within two navigation decisions.
- Prevent dashboard convenience from bypassing evidence, exact-revision, lifecycle, quality-gate, approval, handover or audit rules.

Final product position

QA/QC Manager opens Dashboard -> sees inspections and quality risk -> resolves failed work and corrective verification -> clears governed quality gates -> sees handover blockers -> opens the authoritative quality/handover record -> completes the controlled action. The dashboard is a permission-safe projection, not a second quality database.

2. Goals, Non-Goals and Role Scope

In scope	Out of scope
Exact QA/QC role shell, sidebar and fixed dashboard	A separate quality application that copies Project, Task, Document, Inventory, Asset, Unit or Contract records
Inspection, NCR, CAPA, defect, gate and handover summaries	Automatic visibility of every company project, contractor or confidential issue
My Work, Quality Alerts and actionable exceptions	Arbitrary dashboard drag-and-drop or user-added widgets in v1
Quality planning, ITP, inspection and verification navigation	Replacing Work Progress as physical completion/quantity authority
Handover readiness and package navigation	Replacing Contracts for legal completion or Finance for postings/cost accounting
Workflow approval projections where configured	Workflow ownership of inspection/NCR/defect/handover business state
Responsive mobile and permitted offline drafts	Offline final handover acceptance, ITP publication, NCR closure or stock release

2.1 Role scope model

```
AUTHENTICATED USER
-> Active Company
  -> Active Role: QA/QC Manager
    -> Authorized Project / Contract / Location Scope
      -> Optional Contractor / Discipline / Date Filters
        -> QA/QC Dashboard / Shared Source Pages
```

- Project/company/location/contractor/record scope is enforced server-side.
- Filters narrow authorized data; they never expand access.
- Commercial impact, confidential NCR, client acceptance and restricted documents remain separately permission controlled.
- Multi-role users remain in active QA/QC role until they explicitly switch roles.

3. Application Shell, Header and Exact Sidebar

3.1 Persistent header

```
QA/QC  
[ Company v ] [ Project v ] [ Contractor v ] [ Discipline v ] [ Location v ] [ Date v ]  
[ Global Search ] [ + New ] [ Notifications ] [ Profile / Role ]
```

- Company/project changes re-resolve authorization, dashboard aggregates, saved views and sidebar visibility.
- Contractor, discipline, location and date are narrowing filters only.
- Global Search is permission-safe and opens source records inside the active QA/QC shell.
- Notifications deep-link to source records or Workflow work items without changing active role.

3.2 Exact v1 sidebar

```
QA/QC  
  Dashboard  
  My Work  
  Quality Alerts  
  
QUALITY PLANNING  
  Quality Plans  
  Inspection & Test Plans  
  Templates & Checklists  
  
INSPECTIONS  
  Inspection Requests  
  Inspection Records  
  Material Inspections  
  Work Inspections  
  Tests & Samples  
  Quality Releases  
  
QUALITY ISSUES  
  NCRs  
  Corrective Actions  
  Defects  
  Snag Lists  
  Punch Lists  
  
HANDOVER  
  Handover Dashboard  
  Handover Packages  
  Readiness Matrix  
  Joint Inspections  
  Commissioning  
  Units Ready for Handover  
  Assets Ready for Transfer  
  Defects Liability  
  Final Closeout  
  
REPORTING  
  Reports  
  
SETTINGS  
  Settings
```

- Group labels are visual separators, not routes.
- Unauthorized or configuration-disabled items disappear with clean reflow.
- Tasks, Documents, Projects, Inventory, Assets, Units, Contracts, Calendar and Workflow are linked shared surfaces, not duplicate QA/QC sidebar modules.
- Opening any authorized shared record preserves QA/QC shell and role context.

4. Full QA/QC Dashboard Architecture Map



Layout lock

These dashboard regions are the complete v1 QA/QC Dashboard. There is no generic company KPI area, Finance panel, Sales panel, employee-performance scoring zone, arbitrary chart area or user-added widget region. Deeper analytics live in Reports and authoritative source records.

5. Primary Cues - Exact Set

Cue	Definition	Click-through
Inspections Due	Authorized inspection requests/attempts due today or overdue according to canonical schedule, notice and current state.	Inspection Requests filtered to due/overdue.
Open NCRs	Authorized NCRs not in final Closed state. Critical and overdue state remains visible.	NCR register filtered to open.
Open Defects	Authorized defects/snags/punch items not verified/closed.	Quality Issues filtered to open.
Overdue Corrective Actions	Authorized corrective actions past governed due date and not completed/verified.	Corrective Actions filtered to overdue.
Blocking Quality Gates	Active governed gates preventing continuation, concealment, commissioning, progress or handover.	Quality Gates filtered to blocking.
Handover Blockers	Mandatory unresolved readiness requirements preventing package/unit/asset handover progression.	Handover Readiness filtered to blockers.

- Each cue shows current count, semantic status and drill-through; severity is shown where supported.
- Counts reconcile exactly with the resulting filtered authoritative register.
- No manually editable KPI values or cosmetic readiness percentages are allowed.
- When a readiness score exists, mandatory blockers always take precedence over the score.

6. My Work and Needs My Attention

Work group	Examples	Source / action
Inspections	Due inspection, not-ready visit, failed checkpoint, reinspection required, witness action.	Inspection Request / Inspection Attempt.
NCR	Critical NCR, response due, containment missing, root-cause review, verification required.	NCR.
CAPA	Corrective action overdue, ready for verification, effectiveness failed.	Corrective Action / Task link.
Defects	Defect ready for review, verification rejected, reopened item, punch item due.	Defect / Snag / Punch.
Quality gates	Gate blocking work, missing required evidence/document/test, authorized override review.	Quality Gate / source requirement.
Handover	Missing mandatory requirement, package review, joint inspection, acceptance reservation, final-closeout blocker.	Handover Package / Readiness / Acceptance.
Workflow / execution	Assigned approval/review, linked task, meeting action or document response.	Workflow / Tasks / Calendar / Documents.

6.1 Priority ordering

- Critical safety-linked quality failures, blocking gates and imminent handover blockers appear before routine work.
- Overdue items appear before future-due items unless configured risk priority is higher.
- Items waiting on the user are separated from items waiting on another party.
- Each row includes project, location, record type/code, contractor/party, severity, due state, current stage and required action.

6.2 My Work views

```
MY WORK
+-- My Priority
+-- Due Today
+-- Overdue
+-- Ready for Verification
+-- Waiting on Others
`-- Recently Completed
```

7. Inspection Control

INSPECTION CONTROL

+-- Open Requests

+-- Scheduled

+-- Due Today

+-- Overdue

+-- Ready

+-- In Inspection

+-- Passed

+-- Failed

+-- Reinspection Required

+-- Closed

7.1 Canonical request lifecycle

Draft -> Submitted -> Scheduled -> Ready -> In Inspection -> Result Recorded -> Closed

7.2 Canonical inspection-attempt lifecycle

Started -> Evidence Captured -> Result Recorded -> Verified -> Final

- Failed historical attempts remain preserved; a reinspection is a linked new attempt, never an overwrite.
- Inspection state is rendered from the authoritative QA/QC source. The dashboard never invents a parallel status.
- Due/overdue derives from source date, notice and state, not from a manual dashboard flag.
- Not-ready visits and prerequisite failures remain visible and traceable.

8. Quality Planning, Quality Plans and ITPs

```
Company Quality Framework
-> Project Quality Plan
  -> Discipline / Package Quality Plan
    -> Inspection & Test Plan (ITP)
      -> Activity / Stage
        -> Checkpoint / Acceptance Criterion
          -> Inspection Request / Test / Quality Release
```

8.1 Quality Plan lifecycle

Draft -> Internal Review -> Issued -> Active -> Superseded -> Archived

8.2 ITP checkpoint types

Type	Purpose
Hold	Work cannot continue past the checkpoint until required acceptance/release is recorded.
Witness	Named party is invited/required to witness according to notice and attendance rules.
Surveillance	Observation/control checkpoint without full hold unless configured result triggers a gate.
Review	Review of evidence, documents, records or method compliance.
Test	Measured or functional test with defined acceptance criteria.
Custom	Configured checkpoint type governed by template/version rules.

- Issued plan/ITP revisions are immutable; later changes create a new revision and impact review.
- Every active plan retains exact references to controlled drawings, specifications, methods and contract requirements.
- Changing an active plan/ITP must assess impact on open inspections, tests, releases and handover requirements.

9. Inspection Workspace and Result Integrity

```
INSPECTION
+-- Overview
+-- Scope / Location
+-- ITP Checkpoint
+-- Checklist
+-- Measurements
+-- Evidence
+-- Participants
+-- Documents
+-- Result
+-- Quality Release
+-- Defects / NCRs
+-- Reinspection
`-- Activity
```

9.1 Required decision evidence

Element	Requirement
Scope	Exact project/work package/system/unit/asset/location scope being inspected.
Criterion	Exact ITP/checklist criterion and historical revision used.
Evidence	Required photos, measurements, documents, certificates, notes or signatures.
Actor	Inspector/reviewer/witness identity according to authorization.
Effective time	Inspection/result time distinct from later sync or record update time where relevant.
Result	Configured outcome only after mandatory criteria/evidence validation.

Record integrity rule

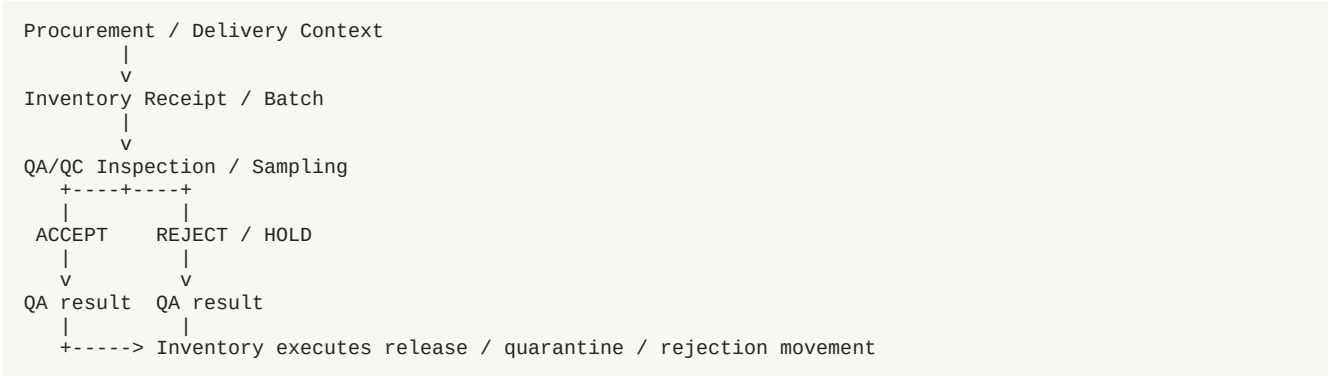
Every official quality decision must expose its exact scope, location, acceptance criterion, evidence, actor and effective time. A final inspection result is never silently rewritten. Corrections use controlled amendment, superseding attempt or other source-approved pattern.

10. Tests, Samples and Material Quality

10.1 Tests and samples

TEST / SAMPLE
+-- Plan / Requirement
+-- Sample Identity
+-- Location / Batch / Asset / System
+-- Measurements
+-- Units / Tolerances / Limits
+-- Laboratory / Certificate
+-- Result
+-- Retest Link
`-- Activity

10.2 Material quality control



- QA/QC owns the material-quality decision and evidence; Inventory owns stock location, quantity and movement.
- Quarantine/release status shown in QA/QC is a source-linked projection and cannot post Inventory movement directly without the owning command path.
- Pending tests and supplier/receipt context may be displayed only with authorized source access.

11. NCR Control

```
NCR
+-- Open
+-- Critical
+-- Overdue
+-- Under Containment
+-- Root Cause
+-- Corrective Plan
+-- Implementation
+-- Verification
+-- Reopened
`-- Closed
```

11.1 Canonical NCR lifecycle

```
Draft -> Issued -> Containment -> Root Cause -> Corrective Plan -> Implementation -> Verification -> Closed
```

11.2 NCR record

```
NCR RECORD
+-- Description / Non-Conformance
+-- Requirement / Violation
+-- Project / Location / Scope
+-- Contractor / Supplier
+-- Severity / Priority
+-- Evidence
+-- Immediate Containment
+-- Disposition
+-- Root Cause
+-- Corrective Actions
+-- Preventive Actions
+-- Verification / Effectiveness
+-- Reinspection
+-- Documents
`-- Activity
```

- Closed NCRs cannot be edited or deleted in place.
- Reopening, supersession, correction and disposition changes must preserve prior state and reason.
- Critical/overdue flags derive from governed severity, due dates and lifecycle.
- Confidential NCR fields are permission filtered independently from general QA/QC access.

12. CAPA and Corrective Work Separation

CAPA
+-- Open Actions
+-- Assigned
+-- In Progress
+-- Overdue
+-- Ready for Verification
+-- Effectiveness Failed
`-- Closed

12.1 Ownership separation

Capability	Owner	QA/QC behavior
Required corrective outcome	QA/QC	Defines required condition/evidence and verification criteria.
Execution task	Tasks / Work Progress	Owens assignee, execution steps, task status and collaboration.
Verification	QA/QC	Confirms correction against exact criterion and evidence.
Reinspection	QA/QC	Creates linked inspection attempt where inspection is required.
Closure	QA/QC	Closes quality issue only after governed validation.

No duplicate task state

The QA/QC dashboard may show linked task state but must not create an independent editable copy of the task lifecycle. A task completing does not automatically close an NCR/defect; QA/QC verification remains separate.

13. Defects, Snags, Punch Lists and Reinspection

QUALITY ISSUES
+-- Defects
+-- Snag Lists
`-- Punch Lists

13.1 Canonical issue lifecycle

Open -> Assigned -> In Correction -> Ready for Review -> Verified -> Closed

13.2 Dashboard views

Open
Overdue
Critical
By Contractor
By Location
Ready for Verification
Verification Rejected
Reopened
Warranty Defects

- Issue records retain exact location, category, responsible party, evidence, source links and verification history.
- Verification rejection returns the issue to a governed correction state without deleting prior evidence.
- Reinspection links the original failure/issue to the new attempt; it does not erase the original result.
- Bulk close is prohibited unless each item passes individual closure validation and returns item-level results.

14. Quality Gates and Blocking Logic

```
QUALITY GATES
+-- Work Continuation
+-- Concealment
+-- Progress
+-- Commissioning
`-- Handover
```

14.1 Gate inputs

```
GATE
+-- Required Inspection
+-- Required Test
+-- Required Document
+-- Required Defect Closure
+-- Required NCR Closure
+-- Required Permit / Certificate
`-- Required Acceptance
```

- A configured blocking gate prevents the governed downstream state until mandatory requirements are satisfied or an authorized override exists.
- Gate state is calculated by the backend from authoritative requirement states; frontend previews are not final authority.
- Overrides are explicit, scoped, reasoned, time-aware and auditable; no generic Skip control exists.
- Gate state may project to Work Progress, Commissioning or Handover but QA/QC never rewrites those modules' authoritative completion records.

15. Handover Readiness Dashboard

HANDOVER READINESS

+-- Packages

+-- Mandatory Blockers

+-- Missing Documents

+-- Open Defects

+-- Commissioning Status

+-- Joint Inspections

+-- Units

+-- Assets

`-- Client / Consultant Acceptance

15.1 Readiness states

Not Ready -> Blocked -> In Preparation -> Ready for Review -> Joint Inspection -> Accepted -> Handover Complete -> DLP -> Final Closed

- Readiness is a gate system, not a cosmetic progress percentage.
- Mandatory blocker state always overrides an optional completeness score.
- Every readiness card/blocker deep-links to the exact requirement, inspection, defect, document, permit, test or acceptance record that produced it.
- Handover readiness remains distinct from physical completion, quality acceptance and contractual completion.

16. Handover Package and Joint Inspection

HANDOVER PACKAGE
+-- Overview
+-- Readiness Matrix
+-- Requirements
+-- Documents
+-- Inspections
+-- Tests
+-- Commissioning
+-- Defects / Punch
+-- Joint Inspection
+-- Acceptance
+-- Reservations
+-- Certificates
`-- Activity

16.1 Package lifecycle

Draft -> In Preparation -> Ready Review -> Joint Inspection -> Accepted -> Handover Complete -> DLP -> Final Closed

16.2 Joint inspection

Area	Requirement
Participants	Authorized internal/external participants according to project, contract and future portal/workflow policy.
Checklist	Versioned handover/joint-inspection checklist and mandatory requirements.
Evidence	Photos, documents, measurements, signatures/acknowledgements as configured.
Punch capture	New handover punch items link directly to package/location and remain separate issue records.
Outcome	Accepted, accepted with reservations or returned/not ready according to configured lifecycle.

Acceptance immutability

A signed/final handover acceptance or completion snapshot becomes read-only. Correction requires an addendum or superseding source record; the dashboard cannot edit it in place.

17. Commissioning, Unit/Asset Handover and DLP

17.1 Commissioning

```
SYSTEM / ASSET
-> Prerequisites
-> Pre-Commissioning
-> Planned Test
-> Ready
-> In Test
-> Result
  +-- Pass -> Acceptance / Operational Readiness
  `-- Fail -> Defect / Corrective Work -> Retest
```

17.2 Unit and asset handover projections

Surface	QA/QC / Handover projection	Authoritative owner retained
Unit	Readiness, defects, keys/meters/doc requirements, acceptance projection.	Units / CRM for unit lifecycle and commercial/client identity.
Asset	Commissioning, inspection, handover documents/training/acceptance evidence.	Assets for activation, custody, maintenance and operational lifecycle.
DLP / warranty defect	Warranty defect, due/aging, verification and final closeout dependency.	Handover/QA for defect record; Assets/Contracts retain their own warranty/obligation records.

18. Project QA/QC Workspace

```
PROJECT QUALITY
+-- Overview
+-- Quality Plan
+-- ITPs
+-- Inspection Requests
+-- Inspections
+-- Tests
+-- Material Quality
+-- Quality Releases
+-- NCRs
+-- Corrective Actions
+-- Defects
+-- Snags / Punch
+-- Quality Gates
+-- Handover
+-- Documents
`-- Activity
```

18.1 Overview tab

- Project quality status and current active plan/ITP references.
- Inspection due/overdue/failed/reinspection state.
- Open NCRs, corrective actions and quality issues by severity.
- Blocking gates and next governed milestone.
- Handover readiness and mandatory blockers where the project is in handover scope.
- Recent meaningful quality/handover activity.

One project, one context

The Project QA/QC Workspace is a QA/QC lens over one Project. It does not create duplicate Project, Task, Document, Inventory, Asset, Unit, Contract or Workflow records.

19. Quality Alerts, Upcoming and Recent Activity

19.1 Quality Alerts

```
QUALITY ALERTS
+-- Critical NCR / Failure
+-- Inspection Overdue
+-- Failed Reinspection
+-- Corrective Action Overdue
+-- Blocking Gate
+-- Missing Mandatory Evidence
+-- Expiring / Missing Controlled Requirement
+-- Handover Blocker
`-- Integration / Source Conflict
```

- Alerts are source-derived exceptions, not user-authored status badges.
- A dismissed UI notification does not resolve the source issue.
- Alerts show freshness and source state; integration delay is distinguished from a real zero/no-issue state.

19.2 Upcoming

- Scheduled inspections and witness points.
- Reinspection dates and corrective-action due dates.
- Joint inspections and commissioning tests.
- Handover review/acceptance dates.
- Linked QA/QC-relevant meetings and workflow SLA deadlines.

19.3 Recent Activity

- Inspection submitted/scheduled/result recorded/verified.
- NCR issued, containment/root cause/corrective plan/verification/closure/reopen.
- Defect/snags/punch created, corrected, rejected, verified or closed.
- Quality gate changed, override applied/expired, handover requirement resolved/blocked.
- Package readiness/acceptance/commissioning/DLP state changed.
- Controlled document/template/ITP revision used or superseded.

20. Reports and Settings

20.1 Reports

Report family	Examples
Inspections	Open/due/overdue, pass/fail, first-pass yield, reinspection rate, failed checkpoints, aging.
NCR / CAPA	Open/critical/overdue, aging, closure time, repeat cause, corrective effectiveness.
Defects	Open/overdue, density, closure velocity, contractor/location distribution, verification rejection.
Materials / tests	Pending tests, rejection, quality hold context, supplier/receipt quality evidence.
Gates	Blocking gates by project/package/location and unresolved requirement type.
Handover	Package readiness, mandatory blockers, commissioning, units/assets, punch, DLP/final closeout.

20.2 Settings

- Quality plan and ITP types, numbering, statuses, checkpoint types and notice periods.
- Checklist templates, criteria, tolerances and conditional logic.
- Inspection results, release states, severity, priority, issue types and root-cause categories.
- NCR disposition, closure and effectiveness rules.
- Handover hierarchy labels, package templates, readiness requirements, commissioning stages and DLP rules.
- SLA, notification, signature, retention and override/waiver rules where platform governance permits.

Configuration boundary

Platform Configuration may enable/disable approved pages, sections, widgets, features, fields, rules, reports and actions. It cannot grant permission, bypass a governed lifecycle, change source ownership or introduce unapproved dashboard content.

21. Cross-Module Ownership and Source-of-Truth Boundaries

Domain	Authoritative owner	QA/QC / Dashboard behavior
Physical progress / quantities	Work Progress	Reads executed scope/completion; publishes quality gate/acceptance projection only.
Corrective execution	Tasks / Work Progress	Creates/links execution work; does not duplicate task state.
Files / drawings / certificates	Documents	References exact controlled Document ID/revision; no separate QA file store.
Contracts / completion terms	Contracts	Reads requirements/obligations; does not author legal completion state.
Stock / quarantine movement	Inventory	Publishes quality decision; Inventory performs movement/release/reject.
Purchase order / supplier process	Procurement	Consumes supplier/delivery context; returns quality outcome.
Asset identity / custody	Assets	Links inspections/commissioning/handover; Assets owns lifecycle.
Unit lifecycle	Units / CRM	Projects readiness/defects/acceptance; source owns unit/client/commercial state.
Safety incidents / permits	HSE	Links/escalates safety implication; HSE owns safety record.
Official permits / certificates	Government & Legal	Uses as readiness dependency; source owns official record.
Financial postings	Finance	May show permitted cost-impact reference; never posts journals.
Approval routing	Workflow Engine	Embeds/links workflow status and work item; source module owns final business state.

Permanent ownership rule

Modules may display each other's data, but they must not create independent editable copies of another module's source records. Every cross-module relationship uses stable source references and permission-safe projections.

22. Approvals and Workflow Integration

```
QA/QC SOURCE RECORD
-> Submit / Request Decision
  -> Workflow Selection
    -> Work Items / Participants / SLA
      -> Decisions
        -> Source Finalization Request
          -> QA/QC Source Validation
            -> Source State Transition
              -> Workflow Complete
```

22.1 Candidate QA/QC decisions

- ITP / quality-plan publication where configured.
- Quality gate override or waiver.
- NCR closure or exceptional disposition where configured.
- High-risk defect acceptance or closure exception.
- Handover package acceptance, completion or final-closeout decisions.
- Emergency quality/continuity override with independent post-event review.
- Workflow owns routing, participants, delegation, decisions, SLA and escalation.
- QA/QC/Handover owns authoritative record content, validation and final business transition.
- Workflow approval is not treated as final source state until QA/QC/Handover confirms finalization.
- Separation-of-duty rules may prevent the inspector/creator from being sole final approver for configured cases.

23. Data Ownership, Read Models and Core Records

23.1 Core source entities used by dashboard

Family	Representative authoritative entities
Planning	ProjectQualityPlan, QualityPlanRevision, InspectionTestPlan, ITPRevision, ITPStage, ITPCheckpoint, ChecklistTemplateRevision.
Inspection / tests	InspectionRequest, InspectionAttempt, InspectionResult, InspectionItemResult, QualityRelease, TestSample, TestRecord, TestMeasurement.
Issues	QualityIssue, Defect, SnagList/SnagItem, PunchList/PunchItem, NonConformanceReport, NCRDisposition, RootCauseAnalysis, CorrectiveAction, PreventiveAction, ReinspectionLink, IssueVerification.
Handover	HandoverStructureNode, HandoverPackage, HandoverRequirement, ReadinessSnapshot, JointInspection, HandoverAcceptance, CommissioningSystem/TestPack/Test, DefectsLiabilityPeriod, WarrantyDefect, FinalCloseoutRecord.
Shared	QualityDocumentLink, QualityEvidence, QualityLocationLink, QualitySourceLink, QualityActivityEvent, GateOverride.

23.2 Dashboard read model

- Dashboard summaries are read-only projections derived from authoritative source records and workflow work items.
- Read models include stable source reference, record version/freshness, tenant/company/project scope and permission-safe fields.
- Dashboard read models must never become a writable business store.
- Every count/card/alert retains enough source identifiers to drill to the exact records that produced it.

24. Frontend Architecture and State

24.1 Component architecture

```
QAQCShell
+-- QAQCDashboardPage
|   +-- QAQCPrimaryCues
|   +-- QAQCWorkInbox
|   +-- InspectionControl
|   +-- NCRCAPAStatus
|   +-- QualityIssuesStatus
|   +-- QualityGatesPanel
|   +-- HandoverReadinessPanel
|   +-- QAQCUpcoming
|   `-- QAQCRecentActivity
+-- QAQCProjectWorkspace
+-- InspectionWorkspace
+-- NCRWorkspace
+-- HandoverPackageWorkspace
+-- QualityAlertsPage
`-- QAQCQuickCreate
```

24.2 State rules

- Server state is authoritative; client state stores scope filters, UI preferences and draft form state only.
- Query keys include tenant, company, active role, project, contractor, discipline, location, date and authorization version.
- Official transitions wait for server authority; optimistic UI is limited to reversible drafts.
- Scope change cancels/invalidates stale requests so late responses cannot overwrite the current dashboard context.
- URL-serialized filters and deep links support direct navigation to record/tab/checkpoint/issue/requirement.

25. Logical API and Command Boundaries

Logical capability	Purpose
GET QA/QC Dashboard	Scoped primary cues, work inbox, inspection/NCR/issue/gate/handover summaries, upcoming and activity.
GET My QA/QC Work	Assigned/participating quality and workflow work items.
GET Quality Alerts	Permission-safe exception register.
GET Project Quality Overview	Single-project QA/QC summary and tab counts.
GET Inspection Control	Inspection request/attempt query with canonical source states.
GET Handover Readiness	Package/unit/asset readiness and blocker projection.
GET QA/QC Activity	Permission-safe quality/handover event feed.
Source module commands	Create/update/transition inspection, NCR, defect, release, package, acceptance, etc., using authoritative QA/QC/Handover APIs.
Workflow commands	Submit/withdraw/resubmit/respond/decide only through Workflow APIs where configured.

- Backend derives authenticated identity and authorization; client-supplied user IDs never define authority.
- Mutations require tenant/company/project compatibility, valid lifecycle/version, idempotency where required and audit context.
- Official evidence/document references use stable Document IDs and exact revisions.
- No dashboard endpoint directly posts Inventory movement, Finance journal, Task completion, Asset activation or Unit commercial state.

26. Authorization, Isolation and Sensitive Data

26.1 Authorization

- Every sidebar item, dashboard cue, count, row, field, alert, activity event and deep link is backend permission filtered.
- Frontend hiding is not authorization.
- Project/contract/location/contractor scope is enforced on every query and mutation.
- Viewing a defect/NCR never automatically grants access to confidential Contract, Finance, Client, Document or HSE detail.
- Signed acceptance, commercial-impact fields, confidential NCR and external-party data may require separate capabilities.

26.2 Personal dashboard isolation

```
tenant_id
+ company_id
+ user_id
+ active_role_id
+ project_scope
+ contractor / discipline / location filters
+ authorization_version
```

- Cache/read-model keys include tenant, company, user/effective authorization, role and active scope.
- Role/company/project change invalidates or isolates old projections.
- Revocation removes stale content from dashboard, search, activity and deep links according to platform policy.
- Cross-user, cross-company or cross-tenant leakage is a P0 broken-access-control defect.

27. Responsive, Mobile, Offline and Accessibility

27.1 Responsive behavior

Desktop	Tablet	Mobile
Persistent/collapsible QA/QC sidebar; wide inspection/issue tables; split views; evidence drawers.	Two-column dashboard; collapsible nav; horizontal table scroll only where necessary; touch-safe actions.	Compact cues; Work Inbox first; stacked cards; sticky record header; camera/evidence capture; bottom-sheet filters/actions.

27.2 Offline permitted

- Download assigned inspections where Mobile/Offline policy permits.
- Complete draft checklists and measurements.
- Capture notes, evidence and draft signatures.
- Create observations/defects and draft NCRs.
- Record witness/attendance and queue supported draft commands.

27.3 Offline restricted

- Final handover acceptance.
- ITP publication.
- NCR closure.
- Inventory stock release/quarantine finalization.
- Contract completion and bulk final status transitions.

27.4 Accessibility and motion

- WCAG 2.1 AA minimum platform target; keyboard navigation, visible focus, accessible tables and labelled controls.
- Status is never conveyed by color alone.
- Charts/reports expose textual summaries and underlying data.
- Reduced-motion preference removes non-essential motion while preserving state feedback.

28. Empty, Loading, Error and Conflict States

State	Required behavior
No authorized project	Show scoped empty state; never fall back to all company projects.
No attention items	Show clear all-caught-up state without fabricating zero for unavailable sources.
Loading	Skeletons/placeholders preserve layout; do not show stale counts as current without freshness signal.
Partial source failure	Healthy regions remain usable; failed region shows source unavailable/retry and keeps prior data clearly stale if displayed.
Access revoked	Remove protected content, clear stale client state and block deep-link access.
Version conflict	Explain record/source version changed; require reload/review before retry.
Offline conflict	Do not overwrite newer server result; route to conflict review.
Document revision changed	Warn and require governed revalidation when exact source revision changed during open inspection/workflow.

29. Performance, Events, Audit and Observability

29.1 Performance targets

Area	Target / rule
Dashboard/page	First usable QA/QC dashboard/page under 2.5 seconds p75 under normal broadband/enterprise conditions.
Operational lists	Indexed normal filter response under 800 ms p75 where source architecture target applies.
Large records	Virtualize long punch lists, inspection item tables and activity feeds; lazy-load heavy evidence.
Evidence	Lazy thumbnails, signed URLs, background rendition; original files do not block page load.
Scale	Support millions of inspection item results/defects per tenant and tens of thousands of records per project.

29.2 Event/read-model behavior

- Source events update dashboard aggregates incrementally; reconciliation jobs repair drift.
- Domain events use transactional outbox/idempotent consumers so source state and integration events do not diverge.
- Duplicate event delivery must not create duplicate releases, defects, tasks, handover state changes or Inventory actions.

29.3 Audit

- Actor, timestamp/effective time, previous/new state, reason, source, device and correlation ID.
- Exact template/document/criterion revisions used.
- Overrides, waivers, reopening, supersession and corrections.
- Evidence added/removed, final document generation and sensitive downloads where required.
- No hard deletion of active/historical quality and handover records.

30. Acceptance Criteria

1. QA/QC login opens the fixed QA/QC Dashboard and exact sidebar defined in this PRD.
2. Dashboard contains exactly the approved regions and six primary cues.
3. All counts/cards/alerts are derived from authoritative QA/QC/Handover/Workflow records and reconcile with drill-down.
4. Company/project/contractor/discipline/location/date filters narrow data without expanding authorization.
5. My Work prioritizes assigned inspections, NCR/CAPA, defect verification, gates and handover actions without duplicating source state.
6. Inspection Request and Inspection Attempt lifecycles remain source-controlled and historical failed attempts are preserved.
7. Official inspection results retain exact scope, criterion/template revision, evidence, actor and effective time.
8. Quality Plans and ITP issued revisions are immutable and active-revision changes trigger impact review.
9. Material QA/QC decisions link to Inventory movements without QA/QC owning stock quantity/location.
10. NCRs support containment, root cause, corrective plan, implementation, verification, reopen and immutable closure history.
11. Corrective execution remains in Tasks/Work Progress while QA/QC owns verification and closure.
12. Defect/snags/punch verification and reinspection preserve prior evidence and attempts.
13. Blocking quality gates cannot be bypassed by dashboard UI and authorized overrides remain explicit/auditable.
14. Handover readiness exposes mandatory blockers and never allows a cosmetic score to override them.
15. Handover Package/Joint Inspection/Acceptance preserve exact requirement/evidence/history and final acceptance immutability.
16. Physical completion, quality acceptance, handover readiness and contractual completion remain separate states.
17. Workflow owns routing/decisions while QA/QC/Handover owns final source state and validation.
18. Opening shared Project/Task/Document/Inventory/Asset/Unit/Contract/Calendar/Workflow content preserves QA/QC shell.
19. Backend permissions protect every field/count/link/activity; no linked-module access expansion occurs.
20. Cross-user/company/tenant negative isolation tests pass.
21. Responsive/mobile/offline rules behave as defined and restricted final actions cannot complete offline.
22. Loading, empty, partial failure, revocation, stale revision and conflict states are explicit and safe.
23. Performance, large-data, accessibility and visual-regression targets pass.
24. No unapproved dashboard widget or sidebar item is present.

31. Explicitly Not Allowed and Definition of Done

31.1 Explicitly not allowed

- Additional QA/QC dashboard KPI cards, charts, modules or shortcuts not defined in this PRD.
- User rearrangement of dashboard regions in v1.
- Automatic access to every company project, contractor, supplier, client or confidential issue.
- Finance, Sales, HR or unrelated company KPI widgets on the QA/QC Dashboard.
- Duplicating Project, Task, Document, Inventory, Procurement, Asset, Unit, Contract, HSE, Finance or Workflow records in dashboard storage.
- Editing final inspection results, closed NCRs, verified defects or signed handover acceptance in place.
- Treating task completion as QA/QC verification/closure.
- Posting stock release/quarantine, Finance journals or Asset/Unit lifecycle transitions directly from dashboard-owned state.
- Using readiness percentage to conceal mandatory blockers.
- Final handover acceptance, ITP publication or NCR closure offline.
- AI defect judgment, automated acceptance, external lab/BIM/CDE/IoT integrations or other future-roadmap capability in v1 unless separately approved.
- Hard-coded role checks replacing platform capability/scope authorization.

31.2 Definition of Done

1. Exact QA/QC sidebar, header and fixed dashboard layout are implemented.
2. Six primary cues, Work Inbox, inspection/NCR/issue/gate/handover regions reconcile with authoritative sources.
3. Quality planning, inspection, NCR/CAPA, defect/reinspection, gate and handover deep links open exact source records.
4. Project QA/QC Workspace and Handover Package workspace preserve context and source ownership.
5. Workflow integration passes submit/decision/source-finalization and separation-of-duty tests.
6. Role-context persistence and permission filtering pass direct-link/search/notification/shared-route tests.
7. Responsive/mobile/offline/accessibility/loading/error/conflict states pass visual and functional QA.
8. Performance and cache-isolation targets pass with representative enterprise data.
9. Audit, immutable-history and negative isolation tests pass with zero critical defects.
10. Dashboard scope lock is respected with zero unapproved elements.

Final implementation rule

If an element is not defined in PRD_QAQC_Dashboard, it does not appear on the QA/QC Dashboard or QA/QC role sidebar. New dashboard content requires an approved PRD revision.

Appendix A. Source Alignment Matrix

PRD decision	Primary source alignment
Quality/Handover information architecture	PRD_QAQC_Handover + QA/QC & Handover Full Architecture.
Dashboard inspection/NCR/defect/material/commissioning/handover metrics	PRD_QAQC_Handover Quality Dashboard + Full Architecture dashboards/analytics.
Quality plans, ITPs, checkpoints and exact revisions	QA/QC & Handover Full Architecture planning/ITP architecture.
Inspection lifecycle, attempts, evidence and releases	QA/QC & Handover Full Architecture inspection/test lifecycle and integrity rules.
NCR/CAPA, defects, snag/punch and reinspection	QA/QC & Handover Full Architecture issue architecture and lifecycle.
Quality gates and blocker precedence	QA/QC & Handover Full Architecture quality gate/readiness rules.
Handover package/readiness/joint inspection/commissioning/DLP	QA/QC & Handover Full Architecture Handover architecture.
Cross-module ownership	QA/QC source PRDs: Work Progress, Tasks, Documents, Inventory, Procurement, Assets, Units, Contracts, HSE, Government & Legal, Finance boundaries.
Approval routing and source finalization	Approvals & Workflow Engine v1.0.
Role authority and isolation	Complete Role & Permission Matrix v1.0.
Shared role shell behavior	Role Context and Shared Module Navigation.
Fixed role dashboard, progressive disclosure, responsive/accessibility	Platform UI/UX Architecture and this PRD.

Appendix B. Route and Context Map

```

/quality/dashboard
-> /quality/my-work
-> /quality/alerts
-> /quality/plans / itps / templates
-> /quality/inspection-requests / inspections / tests / releases
-> /quality/ncrs / corrective-actions / defects / snags / punch
-> /handover/dashboard / packages / readiness / commissioning / dlp / closeout
-> /quality/reports / settings

```

```

Shared source routes
-> Project / Task / Document / Inventory / Procurement / Asset / Unit / Contract / HSE / Calendar /
Workflow
-> active QA/QC role shell remains resolved from session context

```

Final architectural position

Work Progress proves that work was executed. QA/QC proves that it was acceptable. Handover proves that the completed result, its documentation and its responsibilities were transferred correctly. The QA/QC Dashboard presents those authoritative states without collapsing them into one generic completion percentage or duplicating connected-domain ownership.